Grant Achuzia

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EDUCATION

Carleton University ***September 2021 – April 2026***
Bachelor of Engineering in Computer Systems Engineering Ottawa, ON

EXPERIENCE

Lumentum ***January 2025 – August 2025***
System Software Solution Test Intern | ***Design Verification & Test Intern*** *Ottawa, ON*

- Built end-to-end TypeScript test automation and CI/CD pipelines to cut manual regression cycles from hours to under 30 minutes per release via GitHub Actions + Selenium/Tauri E2E flows.
- Wrote Bash and SSH/gNOI/gNOIC scripts to automate firmware upgrades & connectivity checks across networked devices, reducing routine engineer intervention.
- Developed a Python CLI with a plugin-based parse–render–export pipeline to auto-generate PPTX/PDF verification reports from device logs, standardizing reporting and speeding failure triage.

Maintenance Drone Co. ***May 2024 – December 2024***
Software Developer Intern *Ottawa, ON*

- Built a Python tool using Shapely library and Matplotlib that converts KML/XML map files into drone waypoint flight paths for inspection missions.
- Reduced point-cloud processing time by implementing NumPy grid downsampling and noise filters, keeping enough detail for accurate 3D model annotation in Blender.

Roof Maintenance Solutions ***May 2022 – December 2023***
Software Developer Intern *Ottawa, ON*

- Wrote Python scripts to import and export roof inspection annotation data between different formats, making report generation faster.
- Used regex patterns to automatically find and fix errors in survey XML files, preventing incorrect reports that would require re-inspections.

SKILLS

Languages: Python, C, C++, C#, JavaScript, TypeScript, Java, SQL, Bash, PowerShell
Frameworks & Tools: PyTorch, Flask, React, React Native, Docker, Git, GitHub Actions, Firebase, Linux, Jira, Apps Script
Concepts: Machine Learning, Data Structures & Algorithms, CI/CD, DevOps, Agile/Scrum, Test Automation

PROJECTS

CU-Bytes
Engineering Capstone | ***PyTorch, Flask, React Native/Expo, SQLite***

- Trained and deployed a ResNet50-based PyTorch model that identifies 100+ campus dining items from photos, including the data pipeline, model training, and a Flask API used by the mobile app.
- Implemented end-to-end ML integration in a React Native/Expo app, including image capture/upload flows, real-time prediction UI, and performance optimizations for network usage and UI responsiveness.

Cell Reminders
Google Sheets Add-on | ***Apps Script, JavaScript, HTML/CSS, Google Calendar/Sheets APIs***

- Created a Google Sheets add-on that syncs events and tasks between selected cells and Google Calendar, allowing users to create calendar entries directly from spreadsheet data.
- Implemented a sidebar interface with separate tabs for events and tasks, cell selection detection, form validation, and automatic updates when cells change.

Fire Alarm Notification System (FANS)
Engineering Project | ***Python, Flask, JavaScript, Firebase, Raspberry Pi***

- Designed a real-time web dashboard that visualizes fire alarm status & supports remote configuration of alarm parameters.
- Connected Raspberry Pi Pico sensors to LEDs and buzzer hardware, and synchronized alarm events with Firebase so the dashboard updates immediately when alarms trigger.

ACTIVITIES

Licensed Drone Pilot — Transport Canada Small RPAS Advanced Operations certified (PC2214005330).

2024 Canadian Engineering Competition — National Programming Category (Bronze Medalist) in Calgary, AB.